Evaluation Execution Guide — L2 & L3

Assuming Models Are Ready

> **TL;DR** Models are trained. This document gives the step-by-step order for **L2 (Representation Quality)** and **L3 (Structural Coherence)** evaluation.
>
> L1 (RAG evaluation) is in a separate document.
>
> Paper basis per step:
>
> - **Step 1 (L2):** *Graph-level Representation Learning with Joint-Embedding Predictive Architectures* (Skenderi et al., 2023)
> - **Step 2 (L3):** Skenderi et al., 2023 + *Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture* (Assran et al., 2023)
> - **Step 3 (Ablation):** Skenderi et al., 2023 + Yang et al., 2026 + Assran et al., 2023

Mapping to Original Execution Plan

Some items from the original plan are included here; others are excluded or relocated. The table below explains each decision.

Original Plan Item	Status	Where	Why
Masked-subgraph accuracy	✅ Included	Step 1 (L2)	Renamed "masked-subgraph prediction error." Directly supported by Graph-JEPA (Skenderi et al., 2023).
Held-out next-state accuracy	❌ Excluded	Appendix	Clin-JEPA models hourly ICU trajectories; our data is discharge summary graphs — a final-state summary, not a temporal sequence. Data mismatch.
Edge-plausibility ranking	❌ Replaced	Appendix	Graph-JEPA evaluates at graph level, not edge level. "Good latent prediction = good edge classifier" is not shown in the paper. Replaced by **graph-level classification** (Step 2, L3), which Graph-JEPA Table 1 directly supports.
Retrieval lift	➡️ Relocated	L1 document	This measures Faithfulness Δ — a RAG metric. Belongs with the other RAG metrics in the L1 document.
Ablations (with vs without)	✅ Included	Step 3	Renamed from "with-physics vs without-physics" to "with vs without Graph-JEPA refinement." Three conditions: No Refinement / Transformer / Graph-JEPA.

Common Prerequisites

Conditions

The RAG pipeline stays the same. Only the graph changes.

Condition	Graph
(A) No Refinement	GLiNER2 raw graph, used as-is
(B) Transformer	GLiNER2 raw graph → Transformer refinement
(C) JEPA	GLiNER2 raw graph → Graph-JEPA refinement

- **Data:** MIMIC-IV-Note discharge summaries
- **Held constant:** embedding model, retriever (top-K), generator (LLM)
- **Varied:** graph refinement method (A, B, C)

Model Outputs Required

Confirm these before starting:

Deliverable	Description	Format
Raw graph	GLiNER2 output from MIMIC-IV-Note	Node/edge list or JSON
Transformer-refined graph	Raw graph after Transformer refinement	Same
JEPA-refined graph	Raw graph after Graph-JEPA refinement	Same
Patient embeddings	Per-patient graph embedding vectors, all 3 conditions	Numpy array or CSV

Check that all three graphs share the same schema and patient set.

Step 1: L2 — Representation Quality

> **Paper basis:**
>
> - *Graph-level Representation Learning with Joint-Embedding Predictive Architectures* (Skenderi et al., 2023; hereafter Graph-JEPA) — masked subgraph prediction in latent space

Input: Trained JEPA/Transformer models + MIMIC graph data

1.1 Masked-Subgraph Prediction — *Graph-JEPA (Skenderi et al., 2023)*

We adapt Graph-JEPA's core task: "predicting the latent representations of masked subgraphs starting from the latent representation of a context subgraph" (Skenderi et al., 2023).

Procedure:

- Mask 20% of the patient-state graph nodes
- The model predicts the masked portion from the remaining context

- "The goal of the network is to predict the latent embeddings of randomly chosen target subgraphs, given one random context subgraph, conditioned on the positional information of each target subgraph." (Skenderi et al., 2023)

A model that has learned clinical structure should predict masked subgraphs better than one that has not.

1.2 Metrics

Masked-subgraph prediction error — *Graph-JEPA (Skenderi et al., 2023)*

The distance between predicted and target latent representations. Graph-JEPA uses Smooth L1: "We use the smooth L1 loss as the distance function, as it is less sensitive to outliers compared to the typical L2 loss." (Skenderi et al., 2023). Cosine distance and L1 distance are also viable.

z_std (collapse check) — *Clin-JEPA: A Multi-Phase Co-Training Framework for Joint-Embedding Predictive Pretraining on EHR Patient Trajectories (Yang et al., 2026; hereafter Clin-JEPA)*

Does the encoder collapse to constant outputs during training? "zstd drops below 0.05 — about 7% of the SFT-pretrained baseline (zstd=0.700) and the operational threshold below which downstream linear probes fail to converge." (Yang et al., 2026, Figure 2b)

- Track z_std throughout training
- z_std < 0.05 → collapse

Training time — *Graph-JEPA (Skenderi et al., 2023)*

Wall-clock time to train, under identical hardware. Graph-JEPA reported that it "can run up to 2.5x faster than Graph-MAE and 8x faster than MVGRL" (Skenderi et al., 2023, Table 4).

1.3 L2 Results Table

Metric	Source	No Refinement	Transformer	JEPA
Masked-subgraph prediction error ↓	Graph-JEPA	N/A		
z_std (collapse check)	Clin-JEPA	N/A		
Training time (sec) ↓	Graph-JEPA	N/A		

No Refinement has no trained predictor, so these metrics do not apply to it. Graph-level classification (Step 2) covers all three conditions.

Step 2: L3 — Structural Coherence

> **Paper basis:**
>
> - *Graph-level Representation Learning with Joint-Embedding Predictive Architectures* (Skenderi et al., 2023) — graph-level classification, latent space structure, distance function ablation

> - *Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture* (Assran et al., 2023; hereafter I-JEPA) — linear probing protocol
> - *Clin-JEPA: A Multi-Phase Co-Training Framework for Joint-Embedding Predictive Pretraining on EHR Patient Trajectories* (Yang et al., 2026) — latent geometry diagnosis

Input: Refined graphs (A, B, C)

2.1 Graph-Level Classification — *Graph-JEPA (Skenderi et al., 2023)*

Graph-JEPA's main evaluation: freeze the learned embeddings, train a simple classifier on top, measure accuracy. "The subgraph representations are pooled to create a vector representation for the whole graph... This vector representation is then used to fit a linear model with L2 regularization for the downstream task. Specifically, we employ Logistic Regression with L2 regularization on the classification datasets." (Skenderi et al., 2023)

Procedure:

1. Pool each patient's graph embedding into one vector per condition (A, B, C)
2. Freeze embeddings — train only a Logistic Regression classifier
3. Classify patient outcomes (e.g., mortality, diagnostic category)
4. 10-fold cross-validation, 5 seeds — Graph-JEPA's protocol: "We report the accuracy of ten-fold cross-validation for all classification experiments over five runs (with different seeds)." (Skenderi et al., 2023)

If JEPA-refined embeddings classify better, the refinement captures more meaningful clinical structure. No edge-level decomposition needed.

2.2 Latent Space Structure — *Graph-JEPA (Skenderi et al., 2023)*

Does JEPA refinement produce more structured embeddings than Transformer refinement? Graph-JEPA showed that "the change in the curvature of the embedding using the Graph-JEPA objective is noticeable" (Skenderi et al., 2023, Figure 3).

- Project graph embeddings to 2D with t-SNE or UMAP, one plot per condition
- Compare: does JEPA show clearer structure?

2.3 Distance Function Ablation — *Graph-JEPA (Skenderi et al., 2023)*

Graph-JEPA compared "different distance functions" including "Euclidean, Hyperbolic, and Ours" (Skenderi et al., 2023, Table 3) and found that "learning the distance between patches in the 2D unit hyperbola provides a simple way to get the advantages of both embedding types." We test the same alternatives for our refinement training:

- **Euclidean** (L2) — baseline
- **Cosine** — scale-invariant
- **Smooth L1** — Graph-JEPA's default
- **Hyperbolic** (Poincaré) — if numerically stable at our embedding dimension

2.4 L3 Results Table

Metric	Source	No Refinement	Transformer	JEPA
Graph-level classification accuracy ↑	Graph-JEPA			
Latent space structure (qualitative)	Graph-JEPA	UMAP/t-SNE	UMAP/t-SNE	UMAP/t-SNE

Note: **Retrieval Lift** (Faithfulness Δ) is a RAG metric, covered in the L1 document.

Step 3: Ablation — With vs Without Graph-JEPA Refinement

> **Paper basis:**
>
> - *Graph-level Representation Learning with Joint-Embedding Predictive Architectures* (Skenderi et al., 2023) — distance function ablation (Table 3)
> - *Clin-JEPA* (Yang et al., 2026) — curriculum ablation (§5.2)
> - *I-JEPA* (Assran et al., 2023) — prediction space ablation (Table 7)

Compare three conditions:

Condition	Pipeline
With Graph-JEPA	GLiNER2 → Graph-JEPA refinement → RAG
Without Graph-JEPA	GLiNER2 → RAG (no refinement)
Transformer baseline	GLiNER2 → Transformer refinement → RAG

Clin-JEPA validated each training component by removing it and measuring the damage: "Without warmup, the encoder collapses during early JEPA refinement... Without alignment, the predictor instead learns to map online-space contexts to target-space outputs." (Yang et al., 2026)

I-JEPA showed that prediction space matters: "The semantic level of the I-JEPA representations degrades significantly when the loss is applied in pixel-space, rather than representation space." (Assran et al., 2023, Table 7)

Re-run L1 RAG metrics under all three conditions. Test significance with paired t-test or Wilcoxon signed-rank.

Metric	No Refinement	Transformer	Graph-JEPA	p-value
Faithfulness				
Answer Relevance				
Context Relevance				

Step 4: Compile Tables & Figures

1. Merge Steps 1–3 into paper-ready tables
2. Figures:
    - UMAP / t-SNE (Step 2.2)
    - Bar charts: A vs B vs C
    - Ablation chart: with vs without Graph-JEPA
3. Significance tests on all comparisons
4. Results narrative

Execution Order

```
Prerequisites   Receive model outputs, verify formats
                   ↓
Step 1           L2: Representation quality
                   ↓
Step 2           L3: Structural coherence
                   ↓
Step 3           Ablation: with vs without Graph–JEPA
                   ↓
Step 4           Compile tables, figures, significance tests
```

Steps 1 and 2 can run in parallel. Step 3 needs the L1 RAG infrastructure (see L1 document).

References

Paper	Citation	arXiv	Steps
Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture	Assran et al., 2023 (I-JEPA)	2301.08243	Steps 2, 3
Graph-level Representation Learning with Joint-Embedding Predictive Architectures	Skenderi et al., 2023 (Graph-JEPA)	2309.16014	Steps 1, 2, 3
Clin-JEPA: A Multi-Phase Co-Training Framework for JEPA Pretraining on EHR Patient Trajectories	Yang et al., 2026 (Clin-JEPA)	2605.10840	Steps 1, 3
A Path Towards Autonomous Machine Intelligence	LeCun, 2022	Open Review	Background

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Appendix: Excluded Metrics

Originally Proposed but Removed

These metrics were in earlier drafts but removed. All three share the same problem: **the source papers evaluate representation quality; we were using them to measure graph correctness. That jump is not supported by the papers.**

A reviewer would ask: *"Why does latent prediction quality imply graph correctness?"* We cannot fully answer that.

Metric	**Source**	**Assumption**	**Why Removed**
Edge Plausibility AUROC	Graph-JEPA (Table 2)	latent consistency → edge correctness	Graph-JEPA predicts masked subgraph embeddings, not individual edge labels. "Good latent prediction = good edge classifier" is not shown.
Edge-Level Ranking (Hits@K)	Graph-JEPA	subgraph score → edge score	A low subgraph score may reflect context, not a specific edge. Decomposing subgraph scores to edges is not supported.
Graph-State Transition (next-state F1, rollout drift, latent L1 error)	Clin-JEPA (§3, §5.2)	discharge summary = temporal trajectory	Clin-JEPA models hourly ICU data. Our data is discharge summaries — final-state snapshots, not time series.

From *Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture* (Assran et al., 2023)

Metric	**Description**	**Why Excluded**
1% ImageNet evaluation	"Adapt the pretrained models for ImageNet classification using only 1% of the available labels." (Table 2)	Vision benchmark. We work with clinical graphs.
Transfer learning (CIFAR100, Places205, iNat18)	"Linear-evaluation on downstream image classification tasks." (Table 3)	We evaluate within one domain (MIMIC-IV), not across domains.
Clevr/Count, Clevr/Dist	"Linear-evaluation on downstream low-level tasks consisting of object counting and depth prediction." (Table 4)	Vision-only tasks. No clinical equivalent.
Pixel-space vs representation-space ablation	"The semantic level of the I-JEPA representations degrades significantly when the loss is applied in pixel-space, rather than representation space." (Table 7)	We work in representation space only. No pixel-space equivalent exists for graphs.
Masking strategy ablation	"Comparison of proposed multi-block masking with other masking strategies." (Table 6)	Training design choice, not an evaluation metric.

From *Graph-level Representation Learning with Joint-Embedding Predictive Architectures* (Skenderi et al., 2023)

Metric	Description	Why Excluded
Peptides-func: Test AP	"A multi-label graph classification dataset... 10 labels, imbalanced." (Table 9b)	Molecular function classification. No clinical equivalent.
MLP vs Transformer ablation	"Performance when parametrizing encoders through MLPs vs Transformer encoders." (Table 5)	Model design choice, not evaluation.
METIS vs random partitioning	"Random patches also provide strong performance." (Table 6a)	Training detail, not evaluation.
Node vs patch RWSE ablation	"Node-level RWSE consistently outperforms patch-level." (Table 6b)	Positional encoding choice, not evaluation.
Poincaré distance comparison	"Optimization for Poincaré embeddings in higher dimensions is problematic, as shown by the NaN loss." (Table 3)	We use Graph-JEPA's default objective. Full comparison is a model concern.

From *Clin-JEPA* (Yang et al., 2026)

Metric	Description	Why Excluded
AUROC/AUPRC on 7 EEP tasks	"7 multi-criteria event-prediction tasks at horizons of 8–48 hours." (Table 5)	Needs per-task MLP probes and task labels. Beyond current scope.
AUROC/AUPRC on 8 binary outcomes	"8 admission-anchored binary outcomes." (Table 7)	Same. Needs 8 task labels and probes.
z_hist → z_full lift	"Lift is +0.019 mean AUROC with CLIN-JEPA winning all 8 of 8 outcomes."	Needs the AUROC infrastructure above.
Peak centroid divergence over time	"Maximum divergence of 10.05 UMAP units at hour 66." (Figure 3d)	Needs per-hour embeddings over 72 hours. Our graphs are static, not hourly.
Five-paradigm curriculum ablation	"V-JEPA 2-AC style, SFT baseline, w/o warmup, w/o alignment." (§5.2)	Training curriculum concern, not evaluation.